

MAM Organic Fertilizer

Business Management System

Technical / Developer Documentation

Complete reference for understanding, maintaining,
and extending the system — sheets, formulas, code & connections.

TECH STACK

Google Sheets • Google Apps Script (JavaScript)
Google Forms • Telegram Bot API • Drive API

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1. System Overview & Architecture

This is a complete business-management system for a fertilizer manufacturer (MAM Organic Fertilizer), built entirely inside a single Google Sheets file plus Google Apps Script. There is no external server, database, or hosting — everything runs in Google's ecosystem, which keeps it free, maintainable by one person, and requiring zero infrastructure.

The owner never touches the spreadsheet directly. He enters data through **Google Forms** on his phone. Apps Script moves that data into clean sheets, runs calculations, and pushes **Telegram** notifications. A **Dashboard** and **Monthly Report** present everything visually, and a menu exports **PDF** records to Google Drive.

Data Flow (end to end)

```

[ Owner's phone ]
  | fills a Google Form
  v
[ Google Form ] ---> [ "<X> Responses" tab ] (raw dump, hidden)
                        |
                        | on-form-submit trigger fires
                        v
                    [ onAnyFormSubmit(e) ROUTER ] (FormSync.gs)
                        | checks which Responses tab
                        v
                [ correct handler: onSaleFormSubmit, etc. ]
                        | writes clean row + copies formulas
                        v
    [ Clean data sheet: Sale / Raw Material / Production / ... ]
        |                                     |
        | formulas recalc                     | checkSmartAlerts()
        v                                     v
    [ Dashboard + Monthly Report ]           [ Telegram notifications ]

[ Daily time trigger ] ---> dailyCheck() ---> payment-due + monthly reminders
[ Spreadsheet menu ]   ---> "PDF Reports" ---> PDF files saved to Drive

```

Core Design Principles

- **Single source of truth:** all configurable values (rates, VAT, thresholds, credentials) live in one Settings sheet. Change one cell, the whole system updates.
- **Forms in, never direct entry:** the owner only uses forms, so data is consistent and the sheet formulas never get broken by hand.
- **One router, not many triggers:** a single on-form-submit trigger routes all six forms. (A separate trigger per form fires for every submission and scrambles data — see chapter 13.)
- **Formulas live in row 4:** each data sheet keeps its calculation formulas in row 4 as a template; the script copies them into each new row. Row 4 must always contain valid formulas.

- **Idempotent alerts:** PropertiesService flags prevent the same alert (low stock, ACI batch, 7-day due) from firing repeatedly.
- **Manufacturing stock logic:** finished fertilizer is Produced minus Sold; raw materials are tracked separately and never simply subtracted from sales.

2. Spreadsheet Structure (Tabs & Order)

Tab order matters: the leftmost tab is what opens by default, so Dashboard is first. The six hidden "Responses" tabs are created automatically by the linked Google Forms.

#	Tab	Purpose	Visible?
1	Dashboard	Visual summary — opens first	Yes
2	Sale	All sales records	Yes
3	Production	Fertilizer produced + stock box	Yes
4	Raw Material	Raw material purchases	Yes
5	Investment	Capital / expense records	Yes
6	ACI Company	ACI activity + billing log	Yes
7	Settings	Config & credentials	Yes (grey)
8	Monthly Report	Month-filtered summary	Yes
—	Sale Responses	Raw form dump	Hidden
—	Raw Material Responses	Raw form dump	Hidden
—	Production Responses	Raw form dump	Hidden
—	Investment Responses	Raw form dump	Hidden
—	ACI Responses	Raw form dump	Hidden
—	ACI Bill Responses	Raw form dump	Hidden

The Responses tab names are matched EXACTLY by the router in FormSync.gs. If you rename a Responses tab, you must update the matching string in onAnyFormSubmit, or that form will stop syncing.

3. Data Sheets — Columns & Formulas

Every data sheet follows the same convention: a merged title in row 1, headers in row 3, and data starting at row 4. Formula columns hold an IF-guarded formula in row 4 that the script copies down. Below, formula columns are marked (auto).

3.1 Sale

Col	Header	Source / Formula (in row 4)
A	ID	script-assigned (newRow – 3)
B	Date	form
C	Customer Name	form
D	Address	form
E	Product Name	form (dropdown)
F	Pack Size (kg)	form (1/5/40)
G	Quantity (packs)	form
H	Weight Sold (kg)	=IF(F4="", "", F4*G4)
I	Rate (₹/kg)	form
J	Total Price (₹)	=IF(H4="", "", H4*I4)
K	Collection (₹)	form
L	Due (₹)	=IF(J4="", "", J4-K4)

3.2 Raw Material

Col	Header	Source / Formula (in row 4)
A	ID	script-assigned
B	Date	form
C	Supplier Name	form
D	Product Name	form (dropdown)
E	Bag Weight (kg)	form
F	No. of Bags	form
G	Total Weight (kg)	=IF(E4="", "", E4*F4)
H	Total Amount (₹)	form
I	Payment (₹)	form
J	Due (₹)	=IF(H4="", "", H4-I4)

3.3 Production

Col	Header	Source / Formula (in row 4)
A	ID	script-assigned
B	Date	form
C	Bags Produced (40kg)	form
D	Total Weight (kg)	=IF(C4="", "", C4*40)
E	Note	form (optional)

Fertilizer Stock box (on Production, around G3:H8): tracks finished-goods inventory.

Cell	Label	Formula
H4	Total Produced	=SUM(Production!D4:D1000)
H5	Total Sold	=SUM(Sale!H4:H1000)+SUM('ACI Company'!D4:D1000)*40
H6	Available Stock	=H4 - H5
H7	Available Bags	=ROUNDDOWN(H6/40,0)
H8	Status	=IF(H6<=Settings!B8, "⚠️ LOW STOCK", "✅ STOCK OK")

H5 is the critical link: finished stock is reduced by BOTH normal sales (Sale!H, already in kg) AND ACI deliveries ('ACI Company'!D = delivered bags × 40). This is because ACI packets are filled from the owner's own fertilizer.

3.4 Investment

Col	Header	Source
A	ID	script-assigned
B	Date	form
C	Item Name	form
D	Type	form (Current / Permanent)
E	Value (Taka)	form

No formula columns. Type = Current (recurring, e.g. rent) or Permanent (one-time asset, e.g. van/machine). This split matters for interpreting the data, though both are subtracted as expenses in profit.

4. The Settings Sheet (Single Source of Truth)

Every configurable number lives here. Formulas across all sheets and the Apps Script read from these cells, so changing a rate or threshold is a one-cell edit.

Cell	Setting	Value / Formula	Used by
B3	Company Name	MAM Organic Fertilizer	display
B4	ACI Rate (per kg)	6.10	B6
B5	VAT (per kg)	0.20	B6
B6	Net Rate per kg	=ROUND(B4-B5,2) → 5.90	B7
B7	Net Rate per ton	=ROUND(B6*1000,0) → 5900	ACI bills, createBill
B8	Low Stock Limit	20	stock status, checkLowStock
B9	ACI Min Billing (ton)	50	batch logic
B10	ACI Payment Days	45	due dates, createBill
J4	Telegram Bot Token	(secret token)	getTelegramSettings
J5	Owner Chat ID	(numeric id)	sendToAll
J6	Dad Chat ID	(numeric id, may be blank)	sendToAll

The ROUND() on B6 and B7 is deliberate. Without it, 6.10 – 0.20 stores as 5.8999999... (floating-point error), producing bills like ₺294,999.99 instead of ₺295,000. Always keep the ROUND wrappers.

SECURITY: J4 is a live Telegram bot token and J5/J6 are personal chat IDs. Keep the file private. If the token leaks, regenerate it via @BotFather and update J4.

5. ACI Company — Business Logic & Billing

ACI is a special customer with its own rules. The owner fills ACI's empty 40 kg packets with his own fertilizer, delivers them, and bills in 50-ton batches. This sheet has two independent regions: an activity table (A-E) and a billing log (J-O).

Business Rules

- ACI supplies empty 40 kg packets → owner fills them with his own fertilizer → delivers to ACI.
- Because packets are filled from his stock, ACI deliveries reduce Fertilizer Stock (wired via Production!H5).
- **Pricing:** $6.10 \text{ ₺/kg} - 0.20 \text{ VAT} = 5.90 \text{ ₺/kg} = 5,900 \text{ ₺/ton}$ (from Settings B7).
- **Billing threshold:** minimum 50 tons (1,250 bags) per bill = ₺295,000 per batch.
- **Payment terms:** ACI pays within 45 days; the 45-day clock starts at the bill's completion date.

Activity Table (columns A-E)

Col	Header	Notes
A	ID	script-assigned
B	Date	form
C	Activity	Received / Filled / Delivered
D	Quantity (bags)	form
E	Note	form (optional)

ACI Live Summary (near G/H on the sheet) uses SUMIF on the activity table, e.g.:

```
Total Received = SUMIF(C4:C1000,"Received",D4:D1000)
Total Filled   = SUMIF(C4:C1000,"Filled",D4:D1000)
Total Delivered = SUMIF(C4:C1000,"Delivered",D4:D1000)
Delivered (ton) = Total Delivered * 40 / 1000
Bill Batches    = ROUNDDOWN(Delivered_ton / 50, 0)
```

Billing Log (columns J-O)

Col	Header	Source / Formula
J	Bill No.	script (from form) or manual
K	Date Completed	script / manual
L	Bill (₺)	script: $\text{Math.round}(\text{tons} \times \text{Settings!B7})$
M	Payment Status	Pending / Complete (dropdown)
N	Due Date	script: $K + 45 \text{ days}$ (Settings!B10)
O	Days Left	$=\text{IF}(N4="", "", \text{IF}(\text{TRIM}(M4)="Complete", "Paid \checkmark", N4-\text{TODAY}()))$

The TRIM() in the Days Left formula guards against a trailing space in the "Complete" dropdown value. A literal "Complete " (with space) broke both this formula and the profit SUMIF during the build. If you re-add the dropdown, ensure options are exactly "Pending" and "Complete" with no spaces.

6. Dashboard — Cards, Charts & Helper Tables

The Dashboard is presentation-only — every value is a formula reading from the data sheets. Helper tables (used by charts) live far down the sheet (row 100+) and are hidden.

KPI Cards

Card	Formula (core)
Total Sales	=SUM(Sale!J4:J1000)
Customer Due	=SUM(Sale!L4:L1000)
Fertilizer Stock	=Production!H6 (and H7 for bags)
Total Investment	=SUM(Investment!E4:E1000)

Warning Cards

Card	Logic
Low Stock	=IF(Production!H6<=Settings!B8,"⚠️ LOW STOCK","✅ STOCK OK")
ACI Progress	delivered tons vs 50; % to next bill
Supplier Due	=SUM('Raw Material'!J4:J1000)

Estimated Profit Card (M43:P44)

Definition agreed with the client: (Sales + ACI bills marked Complete) – Investment – Raw Material.

```
=SUM(Sale!J4:J1000)
+ SUMIF('ACI Company'!M4:M1000,"Complete",'ACI Company'!L4:L1000)
- SUM(Investment!E4:E1000)
- SUM('Raw Material'!H4:H1000)
```

Note: ACI income counts only bills with status Complete (cash-basis). Conditional formatting turns the card green when positive, red when negative. A negative result is normal in a month with large one-time Permanent investments.

Charts (4)

- Monthly Sales Trend — line chart from a Jan→Dec helper table (SUMPRODUCT by month).
- Sales by Pack Size — pie (1 / 5 / 40 kg split).
- Production vs Sales — bar (produced vs sold vs available).
- Raw Material Breakdown — pie (spend per material).

Auto Tables

Recent Sales (newest 5, self-updating) uses QUERY:

```
=QUERY(Sale!B4:L1000, "SELECT B, C, J, L WHERE B IS NOT NULL ORDER BY B DESC LIMIT 5",  
0)
```

Alerts table shows /  per condition (stock, supplier due, customer due, ACI bill ready) with conditional-format colouring. Keep cells below a QUERY empty or it throws a "would overwrite" error.

7. Monthly Report — Month-Filtered Summary

A standalone sheet that shows one month's totals, selected via a dropdown in B3. It is the sheet that gets exported. It never touches the Dashboard, so it is safe to restyle.

The Month Selector

- **B3** is a dropdown sourced from a helper list in column Z (Z1:Z24 = 1st of each month, Jan 2026 → Dec 2027), formatted as "mmm yyyy".
- **B3 holds a real date** (the 1st of the chosen month); all section formulas filter against it.

The Filtering Pattern (used by every section)

All sections use SUMPRODUCT matching both month AND year, with a blank-date guard to avoid errors:

```
=SUMPRODUCT(
  (TEXT(Sale!$B$4:$B$1000,"yyyy-mm") = TEXT($B$3,"yyyy-mm"))
  * (Sale!J4:J1000)
)
```

IMPORTANT: the system uses the TEXT(date,"yyyy-mm") method, NOT MONTH()/YEAR(). MONTH() throws "parameter 1 value is negative" when the range contains any non-date/blank cell. TEXT() is robust against blanks and stray values. Use the TEXT pattern for any new month-filtered formula.

Sections & Key Cells

Section	Cells	Notes
Sales	A5:B9	Total, Collected, Due, Count
Raw Material	A11:B14	Bought, Paid, Due
Production	D5:E7	Bags, Total kg
Investment	D11:E14	Current, Permanent, Total
ACI	D16:E19	Delivered, bills; E19 = Complete bills this month
Estimated Profit	A21:E22	=B6+E19 - E14 - B12

E19 (ACI billed, this month, Complete only):

```
=SUMPRODUCT(
  (TEXT('ACI Company'!$K$4:$K$1000,"yyyy-mm")=TEXT($B$3,"yyyy-mm"))
  * ('ACI Company'!M4:M1000="Complete")
  * ('ACI Company'!L4:L1000)
)
```

8. Google Forms & the Response Tabs

Six forms feed the system. Each is linked to the spreadsheet, creating a Responses tab. The router reads `e.values` (the submitted row) — so FIELD ORDER in each form must match what the handler expects. `e.values[0]` is always the Timestamp.

Form	Responses Tab	e.values order (after Timestamp)
Sale	Sale Responses	Date, Customer, Address, Product, PackSize, Quantity, Rate, Collection
Raw Material	Raw Material Responses	Date, Supplier, Product, BagWeight, NoOfBags, TotalAmount, Payment
Production	Production Responses	Date, BagsProduced, Note
Investment	Investment Responses	Date, ItemName, Type, Value
ACI	ACI Responses	Date, Activity, Quantity, Note
ACI Bill	ACI Bill Responses	Activity, BillNo, Date, Tons

If you add/reorder a form question, update the `e.values` index comments AND the `setValue` mapping in that form's handler. A mismatch silently writes data to the wrong column.

9. Apps Script — File-by-File Reference

Open via Extensions → Apps Script. The project is organised into logical files. Function names and responsibilities below.

File	Functions	Responsibility
Telegram.gs	getTelegramSettings, sendTelegram, sendToAll, testTelegram, notifyError	Messaging + error email
FormSync.gs	onAnyFormSubmit (router), findNewRow, 6 handlers, createBill, markBillPaid	Form → clean sheet sync + ACI billing
SmartAlerts.gs	checkSmartAlerts, checkLowStock, checkACIBillReady	Condition alerts (run after each submit)
ScheduledAlerts.gs	dailyCheck, checkACIPaymentDue, checkMonthlyReminders	Daily time-based alerts
PDFExport.gs	onOpen (menu), exportSheetForMonth, exportBillingForMonth, createPDFBlob, helpers	PDF menu + Drive export

9.1 The Router (heart of the system)

```
function onAnyFormSubmit(e) {
  try {
    var sheetName = e.range.getSheet().getName();
    if (sheetName === 'Sale Responses') { onSaleFormSubmit(e); }
    else if (sheetName === 'Raw Material Responses') { onRawMaterialFormSubmit(e); }
    else if (sheetName === 'Production Responses') { onProductionFormSubmit(e); }
    else if (sheetName === 'Investment Responses') { onInvestmentFormSubmit(e); }
    else if (sheetName === 'ACI Responses') { onACIFormSubmit(e); }
    else if (sheetName === 'ACI Bill Responses') { onACIBillFormSubmit(e); }
    checkSmartAlerts();
  } catch (err) {
    notifyError('onAnyFormSubmit', err); // emails the developer on any failure
  }
}
```

Why one router: a spreadsheet on-form-submit trigger fires for EVERY linked form. With one handler per form you would get all six firing on every submission. The router inspects `e.range.getSheet().getName()` and dispatches to exactly one handler.

9.2 findNewRow — the row finder

```
function findNewRow(sheet) {
  var ids = sheet.getRange('A4:A1000').getValues();
  var lastDataRow = 3;
  for (var i = 0; i < ids.length; i++) {
```

```

    if (ids[i][0] !== '' && ids[i][0] !== null) { lastDataRow = i + 4; }
  }
  return lastDataRow + 1;    // newId = newRow - 3
}

```

Why not getLastRow(): formulas dragged down to row 1000 make getLastRow() return 1000, so new rows landed at row 1001 (past the data). This loop finds the last REAL id in column A instead. This is the canonical "next row" finder; the billing functions use the same loop on column J.

9.3 A representative handler (Sale)

```

function onSaleFormSubmit(e) {
  var sheet = SpreadsheetApp.getActiveSpreadsheet().getSheetByName('Sale');
  var v = e.values; // [Timestamp, Date, Customer, Address, Product, PackSize, Qty,
Rate, Collection]
  var newRow = findNewRow(sheet);
  var newId = newRow - 3;

  sheet.getRange(newRow, 1).setValue(newId);    // A: ID
  sheet.getRange(newRow, 2).setValue(v[1]);    // B: Date
  sheet.getRange(newRow, 3).setValue(v[2]);    // C: Customer
  sheet.getRange(newRow, 4).setValue(v[3]);    // D: Address
  sheet.getRange(newRow, 5).setValue(v[4]);    // E: Product
  sheet.getRange(newRow, 6).setValue(v[5]);    // F: Pack Size
  sheet.getRange(newRow, 7).setValue(v[6]);    // G: Quantity
  sheet.getRange(newRow, 9).setValue(v[7]);    // I: Rate
  sheet.getRange(newRow, 11).setValue(v[8]);   // K: Collection

  // copy the 3 formulas from row 4 (Weight, Total, Due)
  sheet.getRange('H4').copyTo(sheet.getRange(newRow, 8));
  sheet.getRange('J4').copyTo(sheet.getRange(newRow, 10));
  sheet.getRange('L4').copyTo(sheet.getRange(newRow, 12));

  SpreadsheetApp.flush();
  var total = sheet.getRange(newRow, 10).getValue();
  var due = sheet.getRange(newRow, 12).getValue();
  sendToAll('🛒 <b>New Sale</b>\n' + 'Customer: ' + v[2] + '\n' +
    'Amount: ¢' + total.toLocaleString() + '\n' +
    'Due: ¢' + due.toLocaleString());
}

```

Pattern for all handlers: write form fields to their columns, copyTo() the formula columns from row 4, flush(), read back computed values, send Telegram. The other handlers (Raw Material, Production, Investment, ACI) are identical in shape with their own column maps. Money uses .toLocaleString() for comma formatting.

9.4 ACI Billing — createBill & markBillPaid

```
function onACIBillFormSubmit(e) {
  var v = e.values; // [Timestamp, Activity, BillNo, Date, Tons]
  var activity = v[1], billNo = String(v[2]).trim(), date = v[3], tons = Number(v[4]) ||
50;
  if (activity === 'Bill Submitted')      createBill(aci, settings, billNo, date, tons);
  else if (activity === 'Payment Received') markBillPaid(aci, billNo);
}

// createBill: finds next empty billing row (loop on col J),
// writes J=BillNo, K=Date, L=Math.round(tons*Settings!B7), M='Pending',
// N=Date+Settings!B10 days, O=Days-Left formula (checks M="Complete").
//
// markBillPaid: loops col J for the bill number; if found sets M='Complete'
// and sends a 💰 confirmation; if not found sends a ⚠️ "Bill Not Found".
```

Status string contract: the code writes exactly "Complete" (no space). The Days-Left formula and the Dashboard/Monthly-Report profit formulas all match on "Complete". Keep these in sync if you ever change the wording.

9.5 Smart Alerts (run after every submit)

```
checkSmartAlerts() -> checkLowStock() + checkACIBillReady()

checkLowStock():
  available = Production!H6 ; limit = Settings!B8
  if available <= limit AND flag 'lowStockWarned' != 'yes':
    send ⚠️ alert ; set flag
  else if available > limit: delete flag      // re-arms for next time

checkACIBillReady():
  deliveredBags = sum of ACI col D where col C == 'Delivered'
  batches = floor(deliveredBags*40/1000 / 50)
  if batches > stored 'aciBatchCount':
    send 🚚 "Batch #N ready" ; store new count
```

Idempotency via PropertiesService: flags survive between runs so alerts fire once per event. Low-stock re-arms when stock recovers; ACI fires once per new 50-ton batch.

9.6 Scheduled Alerts (daily timer)

```
dailyCheck() (wrapped in try/catch -> notifyError):
  checkACIPaymentDue():
    for each unpaid bill (M != 'Complete') with a valid Due Date N:
      if daysLeft === 7 and flag 'aciDue7_<billNo>' not set:
        send 🕒 "Payment Due Soon" ; set flag
  checkMonthlyReminders():
```

```
if today is NOT the 1st -> return  
else send 💰 Customer Due (sum Sale!L) and 🇲🇲 Supplier Due (sum RawMaterial!J)
```

9.7 Error email helper

```
function notifyError(functionName, error) {  
  MailApp.sendEmail('sayemafridil10abc@gmail.com',  
    '⚠ MAM Organic Script Error: ' + functionName,  
    'Function: ' + functionName + '\nError: ' + error.toString() +  
    '\nTime: ' + new Date());  
}
```

Both entry points (onAnyFormSubmit and dailyCheck) are wrapped in try/catch that calls notifyError, so any runtime failure emails the developer instead of failing silently.

10. Triggers Configuration

There are EXACTLY TWO triggers. More than this (especially multiple form-submit triggers) will cause data corruption.

Trigger	Function	Type	Event
1	onAnyFormSubmit	From spreadsheet	On form submit
2	dailyCheck	Time-driven	Day timer (e.g. 6-7 am)

Verify in Apps Script → Triggers (clock icon). If you ever see a second "On form submit" trigger (e.g. a leftover onRawMaterialFormSubmit), DELETE it — that exact bug once dumped every form's data into the Raw Material sheet.

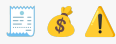
11. Telegram Integration

Notifications use the Telegram Bot API via `UrlFetchApp`. Credentials are read from Settings (J4 token, J5/J6 chat IDs). HTML parse mode allows `<b>` bold.

```
function sendTelegram(chatId, message) {
  var s = getTelegramSettings();
  UrlFetchApp.fetch('https://api.telegram.org/bot' + s.token + '/sendMessage', {
    method: 'post',
    payload: { chat_id: String(chatId), text: message, parse_mode: 'HTML' },
    muteHttpExceptions: true
  });
}
function sendToAll(message) {
  var s = getTelegramSettings();
  if (s.chatIdYou) sendTelegram(s.chatIdYou, message);
  if (s.chatIdDad) sendTelegram(s.chatIdDad, message); // skipped if J6 blank
}
```

Adding a recipient (e.g. the owner / his dad)

- The new person opens Telegram and sends any message to the bot first (required, or Telegram blocks the bot from messaging them).
- Get their numeric chat ID (e.g. via `@userinfobot`).
- Put that ID in Settings J6. `sendToAll` then includes them automatically.

Notification	Trigger point	Emoji
New Sale / Raw Material / Production / Investment / ACI	each handler	
Low Fertilizer Stock	checkLowStock	
ACI Bill Ready (50 t)	checkACIBillReady	
ACI Bill Created / Paid / Not Found	createBill / markBillPaid	
Payment Due Soon (7 days)	checkACIPaymentDue	
Customer / Supplier Due (monthly 1st)	checkMonthlyReminders	

12. PDF Export System

A custom spreadsheet menu (built by onOpen) exports month-filtered PDFs to Google Drive, organised by sheet. ACI activity and ACI billing are exported as separate PDFs.

Menu structure

	PDF Reports	
└	Sale	▸ This Month / Last Month / Choose Month...
└	Raw Material	▸ ...
└	Production	▸ ...
└	Investment	▸ ...
└	ACI Company	▸ ... (activity table, columns A–E only)
└	ACI Billing	▸ ... (billing log J–O, filtered by col K)
└	ALL Sheets	▸ ... (every sheet + ACI Billing)

How it works

- onOpen() builds the menu each time the file opens.
- "Choose Month" prompts for YYYY-MM; This/Last compute from today's date.
- exportSheetForMonth copies the chosen month's rows to a temp sheet (title bar + headers + borders + autosize), converts via the spreadsheet's /export?format=pdf endpoint, saves the blob to Drive, then deletes the temp sheet.
- Folder layout: MAM Organic Records / <Sheet Name> / <Sheet> - <Month Year>.pdf
- ACI Company export is capped to columns A–E; ACI Billing reads J–O and filters by Date Completed (K).

createPDFBlob authenticates with ScriptApp.getOAuthToken() against the export URL (format=pdf, A4, portrait, fitw=true, gridlines=false). If Google changes the export endpoint this is the single function to adjust.

13. Key Bugs Solved (History & Lessons)

These were real problems hit during the build. Documented so they are not reintroduced.

Bug	Cause	Fix
Multiple form triggers scrambled data	One on-form-submit trigger per form fires for ALL forms	Single router + ONE trigger (chapter 10)
New rows landed at row 1001	getLastRow() returns 1000 because formulas filled col to row 1000	findNewRow loops col A for last real ID
Bill showed ₺294,999.99	Float error: $6.10 - 0.20 = 5.8999...$	ROUND() in Settings B6/B7 + Math.round in code
"Cleared" vs "Complete" mismatch	Dropdown used "Complete", code wrote "Cleared"	All code/formulas standardised on "Complete"
Trailing space "Complete " broke status	Dropdown option had a space	TRIM() in Days-Left; clean dropdown option
MONTH() #VALUE!/negative error	MONTH() fails on blank/non-date cells in range	Switched to TEXT(date,"yyyy-mm") matching
Customer Due hugely inflated	Blank Collection → Due = full Total	Always fill Collection (0 if unpaid is wrong; enter actual)
Profit ignored ACI income	SUMIF didn't match "Complete " (space)	Clean status string; profit then correct

14. Maintenance Guide & Future Changes

Common change requests — where to edit

I want to...	Where	How
Change ACI rate or VAT	Settings B4 / B5	Net rate B6/B7 auto-update
Change low-stock threshold	Settings B8	One cell
Change ACI batch size or terms	Settings B9 / B10	One cell
Add a Telegram recipient	Settings J6	Chapter 11 steps
Add a new month to the report	Monthly Report col Z	Add date, format mmm yyyy, extend B3 range
Add a new form field	Form + that handler	Add question; map new e.values index in handler
Add a new data column with a formula	Sheet row 4 + handler	Put formula in row 4; add a copyTo() in handler
Add a new alert	SmartAlerts/Scheduled	New function + a PropertiesService flag

Golden rules when editing

- **Row 4 is sacred.** It holds the formula templates the script copies. Never leave it blank or with wrong references.
- **Keep exactly two triggers.** One form-submit router, one daily timer.
- **Match status strings exactly.** "Pending" / "Complete", no spaces, everywhere (code, formulas, dropdown).
- **Use TEXT(date,"yyyy-mm")** for any month filtering, never MONTH()/YEAR() on ranges that can contain blanks.
- **Responses tab names are contracts.** If you rename one, update the router string.
- **Test after every change** by submitting one of each affected form and watching the sheet + Telegram.

Handover checklist (before giving to the client)

- Clear all test data from every data sheet (keep row 4 formulas).
- Add the owner/dad chat ID to Settings J6.
- Confirm both triggers exist and no extras.
- Confirm Telegram bot token (J4) is valid and the bot has been started by each recipient.
- Protect Settings and formula ranges (right-click tab → Protect) to prevent accidental edits.
- Submit one real entry of each type to verify the full pipeline end-to-end.

15. Quick Reference (Cheat Sheet)

Item	Value
Tech stack	Google Sheets + Apps Script + Forms + Telegram Bot API + Drive
Dedicated account	mamorganic.management@gmail.com
Data starts at	Row 4 (headers row 3, title row 1)
ID rule	newId = newRow - 3
Triggers	onAnyFormSubmit (form submit) + dailyCheck (daily)
Net rate / ton	Settings B7 = $5,900 \div (6.10 - 0.20 \text{ VAT}) \times 1000$
ACI batch	50 tons = 1,250 bags = ₪295,000
ACI terms	45 days (Settings B10)
Stock formula	Produced (Σ Production!D) - Sold (Σ Sale!H + ACI delivered $\times$ 40)
Profit	Sales + ACI(Complete) - Investment - Raw Material
Status strings	Pending / Complete (exact, no spaces)
Month filter	TEXT(date,"yyyy-mm") = TEXT(B3,"yyyy-mm")
Drive PDFs	MAM Organic Records / <Sheet> / <Sheet> - <Month Year>.pdf
Error reports	emailed to sayemafridi110abc@gmail.com via notifyError

End of Technical Documentation

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